

✓ Delivery Performance Audit for Veridi Logistics

Business Problem

Veridi Logistics wants to determine whether customer dissatisfaction is caused by inaccurate delivery estimates and delayed deliveries. This project combines logistics, customer review, and product data to identify the root causes of poor customer experience.

```
# Importing python libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as ns
print('all python labraries loaded succssuccessfully ')
```

```
all python labraries loaded succssuccessfully
```

✓ Story 1: Schema Builder

The objective of this task is to combine orders, customers, and review data into a single master dataset that can be used for integrated logistics and customer satisfaction analysis.

```
# loading the datasets
customers = pd.read_csv('/content/olist_customers_dataset.csv')
reviews = pd.read_csv('/content/olist_order_reviews_dataset.csv')
orders = pd.read_csv('/content/olist_orders_dataset.csv')
products = pd.read_csv('/content/olist_products_dataset.csv')
translation = pd.read_csv('/content/product_category_name_translation.csv')
order_items = pd.read_csv('/content/olist_order_items_dataset.csv')
print('datasets loaded succssuccessfully')
```

```
datasets loaded succssuccessfully
```

✓ Data Exploration

The datasets were explored to understand their structure, identify key variables, inspect data types, and assess data quality before integration and analysis.

```
# exploring the datasets
print('customers:', customers.shape)
print('reviews:', reviews.shape)
print('orders:', orders.shape)
print('products:', products.shape)
print('translation:', translation.shape)
```

```
customers: (99441, 5)
reviews: (99224, 7)
orders: (99441, 8)
products: (32951, 9)
translation: (71, 2)
```

```
# Checking for duplicate rows in each dataset
```

```
print("Customers duplicates:", customers.duplicated().sum())
print("Reviews duplicates:", reviews.duplicated().sum())
print("Orders duplicates:", orders.duplicated().sum())
print("Products duplicates:", products.duplicated().sum())
print("Translation duplicates:", translation.duplicated().sum())
```

```
Customers duplicates: 0
Reviews duplicates: 0
Orders duplicates: 0
Products duplicates: 0
Translation duplicates: 0
```

```
# exploring the datasets
customers.head()
```

	customer_id	customer_unique_id	customer_email
0	06b8999e2fba1a1fbc88172c00ba8bc7	861eff4711a542e4b93843c6dd7febb0	
1	18955e83d337fd6b2def6b18a428ac77	290c77bc529b7ac935b93aa66c333dc3	
2	4e7b3e00288586ebd08712fdd0374a03	060e732b5b29e8181a18229c7b0b2b5e	
3	b2b6027bc5c5109e529d4dc6358b12c3	259dac757896d24d7702b9acbbff3f3c	

```
customers.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99441 entries, 0 to 99440
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id            99441 non-null  object
1   customer_unique_id     99441 non-null  object
```

```
2 customer_zip_code_prefix 99441 non-null int64
3 customer_city            99441 non-null object
4 customer_state           99441 non-null object
dtypes: int64(1), object(4)
memory usage: 3.8+ MB
```

```
reviews.head()
```

	review_id	order_id	review
0	7bc2406110b926393aa56f80a40eba40	73fc7af87114b39712e6da79b0a377eb	
1	80e641a11e56f04c1ad469d5645fdfde	a548910a1c6147796b98fdf73dbeba33	
2	228ce5500dc1d8e020d8d1322874b6f0	f9e4b658b201a9f2ecdecbb34bed034b	
3	e64fb393e7b32834bb789ff8bb30750e	658677c97b385a9be170737859d3511b	
4	f7c4243c7fe1938f181bec41a392bdeb	8e6bfb81e283fa7e4f11123a3fb894f1	

```
reviews.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99224 entries, 0 to 99223
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   review_id                            99224 non-null  object
1   order_id                             99224 non-null  object
2   review_score                         99224 non-null  int64
3   review_comment_title                 11568 non-null  object
4   review_comment_message               40977 non-null  object
5   review_creation_date                 99224 non-null  object
6   review_answer_timestamp               99224 non-null  object
dtypes: int64(1), object(6)
memory usage: 5.3+ MB
```

```
orders.head()
```

	order_id	customer_id	order_s
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	de
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	de
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	de
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	de

```
4 ad21c59c0840e6cb83a9ceb5573f8159 8ab97904e6daea8866dbdbc4fb7aad2c de
```

```
orders.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99441 entries, 0 to 99440
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   order_id                             99441 non-null  object
1   customer_id                         99441 non-null  object
2   order_status                         99441 non-null  object
3   order_purchase_timestamp            99441 non-null  object
4   order_approved_at                   99281 non-null  object
5   order_delivered_carrier_date         97658 non-null  object
6   order_delivered_customer_date       96476 non-null  object
7   order_estimated_delivery_date       99441 non-null  object
dtypes: object(8)
memory usage: 6.1+ MB
```

```
products.head()
```

	product_id	product_category_name	product_name_length
0	1e9e8ef04dbcff4541ed26657ea517e5	perfumaria	40
1	3aa071139cb16b67ca9e5dea641aaa2f	artes	44
2	96bd76ec8810374ed1b65e291975717f	esporte_lazer	40
3	cef67bcfe19066a932b7673e239eb23d	bebes	27
4	9dc1a7de274444849c219cff195d0b71	utilidades_domesticas	37

```
products.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32951 entries, 0 to 32950
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   product_id                             32951 non-null  object
1   product_category_name                  32341 non-null  object
2   product_name_length                   32341 non-null  float64
3   product_description_length            32341 non-null  float64
4   product_photos_qty                    32341 non-null  float64
5   product_weight_g                      32949 non-null  float64
6   product_length_cm                    32949 non-null  float64
7   product_height_cm                    32949 non-null  float64
8   product_width_cm                     32949 non-null  float64
```

```
dtypes: float64(7), object(2)
memory usage: 2.3+ MB
```

```
translation.head()
```

	product_category_name	product_category_name_english
0	beleza_saude	health_beauty
1	informatica_acessorios	computers_accessories
2	automotivo	auto
3	cama_mesa_banho	bed_bath_table
4	moveis_decoracao	furniture_decor

```
translation.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 71 entries, 0 to 70
Data columns (total 2 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   product_category_name                 71 non-null    object
 1   product_category_name_english        71 non-null    object
dtypes: object(2)
memory usage: 1.2+ KB
```

```
# data cleaning
# Converting Order Dates to Datetime
date_columns = [
    'order_purchase_timestamp',
    'order_approved_at',
    'order_delivered_carrier_date',
    'order_delivered_customer_date',
    'order_estimated_delivery_date'
]
for col in date_columns:
    orders[col] = pd.to_datetime(orders[col])
    print("Date columns converted successfully!")
```

```
Date columns converted successfully!
Date columns converted successfully!
Date columns converted successfully!
Date columns converted successfully!
Date columns converted successfully!
```

```
orders.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```

RangeIndex: 99441 entries, 0 to 99440
Data columns (total 8 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   order_id                             99441 non-null  object
 1   customer_id                           99441 non-null  object
 2   order_status                           99441 non-null  object
 3   order_purchase_timestamp               99441 non-null  datetime64[ns]
 4   order_approved_at                      99281 non-null  datetime64[ns]
 5   order_delivered_carrier_date           97658 non-null  datetime64[ns]
 6   order_delivered_customer_date          96476 non-null  datetime64[ns]
 7   order_estimated_delivery_date          99441 non-null  datetime64[ns]
dtypes: datetime64[ns](5), object(3)
memory usage: 6.1+ MB

```

✓ Building the Master Dataset

Orders, customer information, and review data were merged into a single analytical dataset. This integrated dataset serves as the foundation for all subsequent delivery performance and customer satisfaction analyses.

```

## Story 1: Building the Master Dataset
orders_customers = pd.merge(    # merging orders and customers tables
    orders,
    customers,
    on = 'customer_id',
    how = 'left'
)
orders_customers.shape

(99441, 12)

```

```

master_df = pd.merge(    # merging orders_customers and reviews
    orders_customers,
    reviews,
    on = 'order_id',
    how = 'left'
)
master_df.shape

(99992, 18)

```

```
master_df['order_id'].nunique()
```

```
99441
```

```
reviews['order_id'].duplicated().sum()
```

```
np.int64(551)
```

```
# Remove duplicate reviews before merging
reviews_clean = reviews.drop_duplicates(
    subset = 'order_id',
    keep = 'first'
)
reviews_clean.shape
```

```
(98673, 7)
```

```
reviews_clean['order_id'].duplicated().sum()
```

```
np.int64(0)
```

```
master_df = pd.merge(
    orders_customers,
    reviews_clean,
    on = 'order_id',
    how = 'left'
)
master_df.shape
```

```
(99441, 18)
```

```
master_df['order_id'].nunique()
```

```
99441
```

```
master_df['order_id'].duplicated().sum()
```

```
np.int64(0)
```

✓ Story 2: Real Delay Calculator

This analysis calculates the difference between estimated and actual delivery dates and classifies orders into On Time, Late, Super Late, and Not Delivered categories.

```
## Story 2: The "Real" Delay Calculator
#Checking Missing Delivery Dates
master_df['order_delivered_customer_date'].isnull().sum()
```

```
np.int64(2965)
```

```
master_df['order_estimated_delivery_date'].isnull().sum()
```

```
np.int64(0)
```

```
#Creating the Days_Difference Column
master_df['Days_difference'] = (
    master_df['order_estimated_delivery_date'] -
    master_df['order_delivered_customer_date']
).dt.days
```

```
master_df.head()
```

	order_id	customer_id	order_status
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	delivered
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	delivered
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	delivered
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	delivered
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	delivered

```
master_df[['order_estimated_delivery_date',
            'order_delivered_customer_date',
            'Days_difference']].head()
```

	order_estimated_delivery_date	order_delivered_customer_date	Days_difference
0	2017-10-18	2017-10-10 21:25:00	7
1	2018-08-13	2018-08-07 15:27:00	6
2	2018-09-04	2018-08-17 18:06:00	18
3	2017-12-15	2017-12-02 00:28:00	13
4	2018-02-26	2018-02-16 18:17:00	10

```
# Creating Delivery Status
def classify_delivery(days):
    if pd.isna(days):
        return 'Not Delivered'
    elif days >= 0:
        return 'On Time'
```

```

elif days >= -5:
    return 'Late'
else:
    return 'Super Late'
master_df['Delivery_Status'] = master_df['Days_difference'].apply(classify_d

```

```
master_df['Delivery_Status'].value_counts()
```

	count
Delivery_Status	
On Time	88649
Super Late	4212
Late	3615
Not Delivered	2965

```
dtype: int64
```

```
master_df.columns
```

```

Index(['order_id', 'customer_id', 'order_status',
'order_purchase_timestamp',
      'order_approved_at', 'order_delivered_carrier_date',
      'order_delivered_customer_date', 'order_estimated_delivery_date',
      'customer_unique_id', 'customer_zip_code_prefix', 'customer_city',
      'customer_state', 'review_id', 'review_score',
'review_comment_title',
      'review_comment_message', 'review_creation_date',
      'review_answer_timestamp', 'Days_difference', 'Delivery_Status'],
      dtype='object')

```

✓ Story 3: Geographic Analysis

Objective

This analysis identifies the states with the highest percentage of late deliveries to determine whether delivery performance varies geographically.

Business Question

Are certain regions experiencing significantly more delivery delays than others?

```

# We want to group Late and Super Late together
master_df['Late_Flag'] = master_df['Delivery_Status'].isin(

```

```
    ['Late', 'Super Late']
)
```

```
# Calculate Late Deliveries by State
State_Late = master_df.groupby('customer_state')['Late_Flag'].mean()*100
State_Late = State_Late.sort_values(ascending=False)
State_Late.head()
```

	Late_Flag
customer_state	
AL	23.002421
MA	18.875502
PI	15.353535
CE	14.670659
SE	14.571429

dtype: float64

```
# Creating a clean dataset for delivery analysis
analysis_df = master_df[
    master_df['Delivery_Status'] != 'Not Delivered'
]
analysis_df.shape
```

(96476, 21)

```
analysis_df.columns
```

```
Index(['order_id', 'customer_id', 'order_status',
      'order_purchase_timestamp',
      'order_approved_at', 'order_delivered_carrier_date',
      'order_delivered_customer_date', 'order_estimated_delivery_date',
      'customer_unique_id', 'customer_zip_code_prefix', 'customer_city',
      'customer_state', 'review_id', 'review_score',
      'review_comment_title',
      'review_comment_message', 'review_creation_date',
      'review_answer_timestamp', 'Days_difference', 'Delivery_Status',
      'Late_Flag'],
      dtype='object')
```

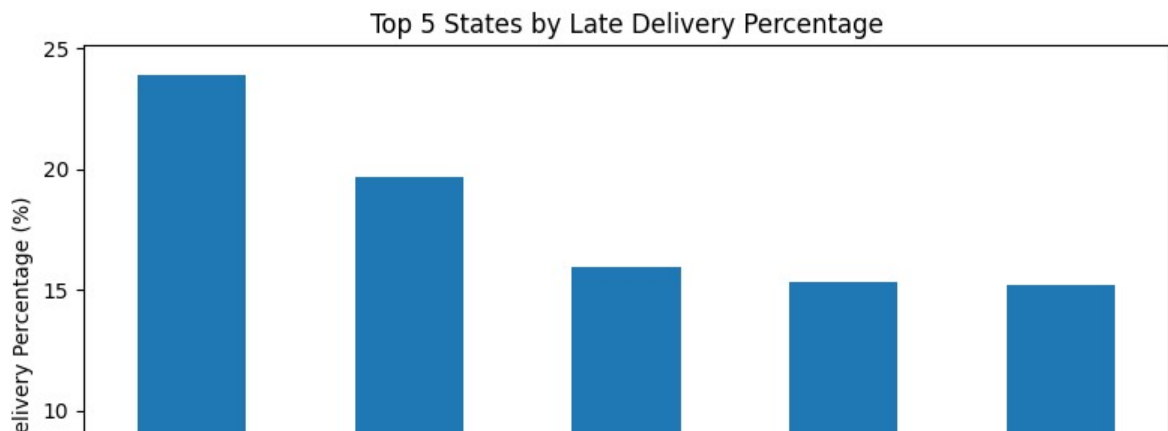
```
# Calculating the % of late orders per State.
State_Late = (
    analysis_df
    .groupby('customer_state')['Late_Flag']
```

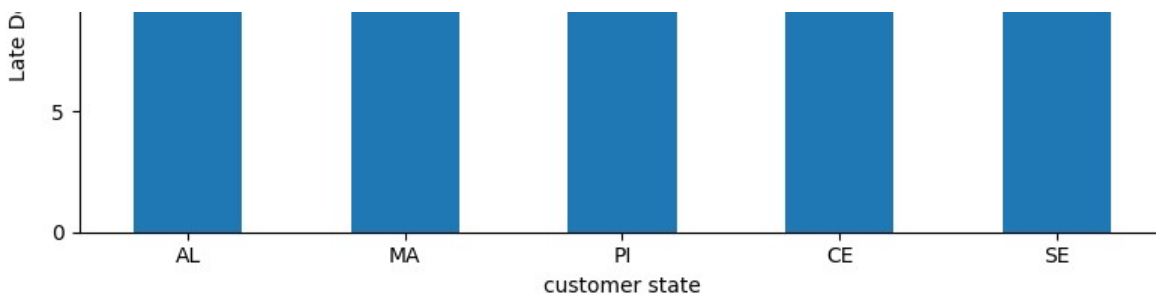
```
.mean()*100  
) .sort_values(ascending = False)  
State_Late.head(10)
```

	Late_Flag
customer_state	
AL	23.929471
MA	19.665272
PI	15.966387
CE	15.324472
SE	15.223881
BA	14.035627
RJ	13.470412
TO	12.773723
PA	12.367865
ES	12.230576

dtype: float64

```
# Creating the Chart  
top10_states = State_Late.head(5)  
plt.figure(figsize=(8,5))  
top10_states.plot(kind = 'bar')  
plt.title('Top 5 States by Late Delivery Percentage')  
plt.xlabel('customer state')  
plt.ylabel('Late Delivery Percentage (%)')  
plt.xticks(rotation=0)  
plt.tight_layout()  
plt.savefig  
plt.show()
```





Insight

States such as AL, MA, and PI recorded the highest late-delivery rates. While the dataset does not contain warehouse or distance information, the concentration of delays in certain regions suggests that geographic and logistical factors may be contributing to delivery performance issues.

✓ Story 4: Sentiment Correlation

This analysis investigates whether delivery delays negatively affect customer review scores.

```
# Comparing Average Review Scores by Delivery Status
master_df.groupby('Delivery_Status')['review_score'].mean()
```

	review_score
Delivery_Status	
Late	3.459361
Not Delivered	1.751671
On Time	4.294358
Super Late	1.786517

dtype: float64

```
Avg_reviews = master_df.groupby('Delivery_Status')['review_score'].mean()
```

```
Avg_reviews.plot(kind = 'bar',  
                  color='#2ca02c')  
plt.title('Average Review Score by Delivery Status')  
plt.xlabel('Delivery Status')  
plt.ylabel('Average Review Score')  
plt.ylim(0,5)  
plt.xticks(rotation = 45)  
plt.show()
```



```
review_comparison = master_df[  
    master_df['Delivery_Status'].isin(['On Time', 'Late'])  
]  
  
review_comparison.groupby('Delivery_Status')['review_score'].mean()
```

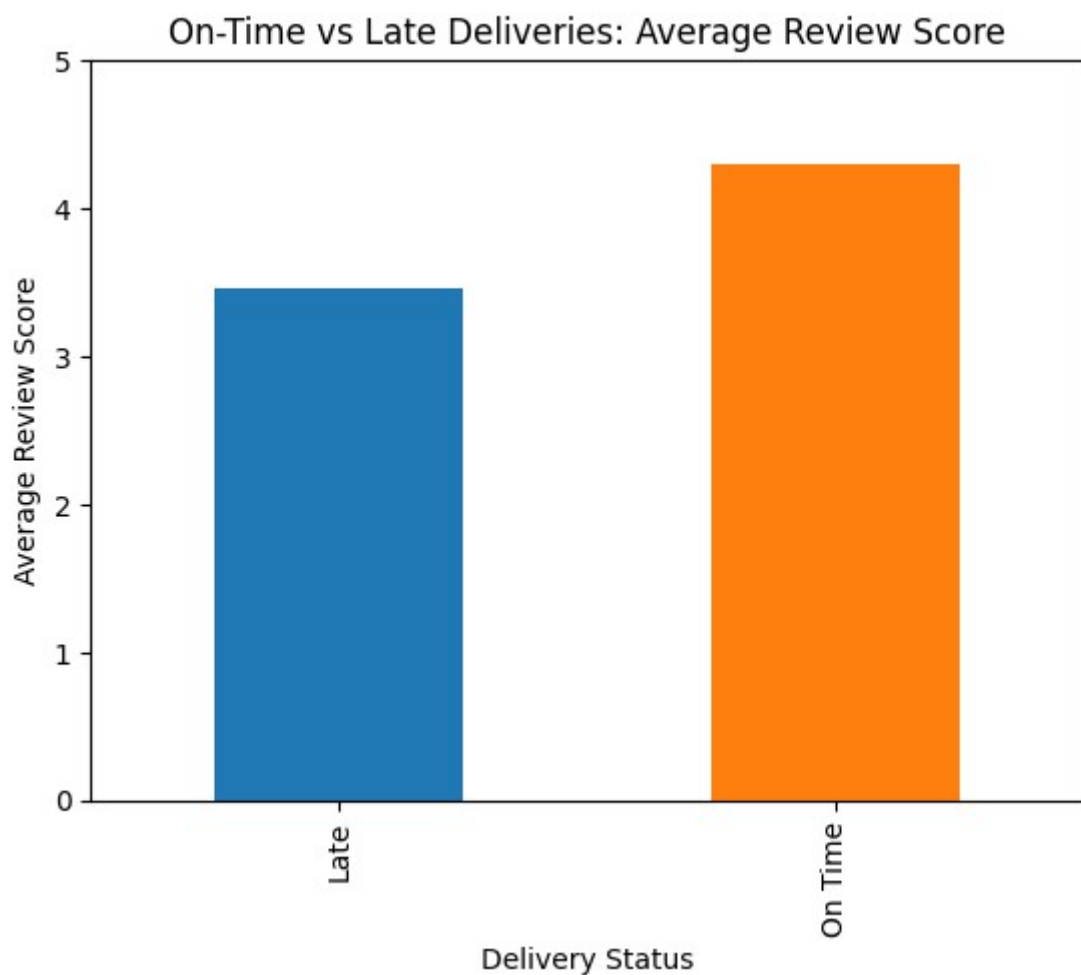
review_score	
Delivery_Status	
Late	3.459361
On Time	4.294358

dtype: float64

```
review_comparison.groupby('Delivery_Status')['review_score'].mean().plot(
    kind='bar',
    color=['#1f77b4', '#ff7f0e']
)

plt.title('On-Time vs Late Deliveries: Average Review Score')
plt.xlabel('Delivery Status')
plt.ylabel('Average Review Score')
plt.ylim(0,5)

plt.show()
```



✓ Story 5: Product Category Translation and Analysis

Product categories are translated from Portuguese to English to support business analysis and identify categories with the highest late-delivery rates.

```
order_items.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 112650 entries, 0 to 112649
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id               112650 non-null  object
1   order_item_id          112650 non-null  int64
2   product_id             112650 non-null  object
3   seller_id              112650 non-null  object
4   shipping_limit_date    112650 non-null  object
5   price                  112650 non-null  float64
6   freight_value          112650 non-null  float64
dtypes: float64(2), int64(1), object(4)
memory usage: 6.0+ MB
```

```
# merging order_items with master_df
master_df = pd.merge(
    master_df,
    order_items,
    on = 'order_id',
    how = 'left'
)
master_df.shape
```

```
(113425, 27)
```

```
master_df['order_id'].nunique()
```

```
99441
```

```
order_items['order_id'].nunique()
```

```
98666
```

```
# Merge Products tables
master_df = pd.merge(
    master_df,
    products,
    on = 'product_id',
    how = 'left'
)
master_df.shape
```

```
(113425, 35)
```

```
# merging translation table
master_df = pd.merge(
    master_df,
```

```
translation,  
on='product_category_name',  
how='left'  
)  
  
master_df.shape
```

```
(113425, 36)
```

```
master_df[  
    ['product_category_name',  
     'product_category_name_english']  
].head()
```

	product_category_name	product_category_name_english
0	utilidades_domesticas	housewares
1	perfumaria	perfumery
2	automotivo	auto
3	pet_shop	pet_shop
4	papelaria	stationery

```
# Calculating Late Delivery Rate by Product Category  
category_late_rate = (  
    master_df.groupby('product_category_name_english')['Late_Flag']  
    .mean()  
    .sort_values(ascending= False)*100  
)  
category_late_rate.head(10)
```

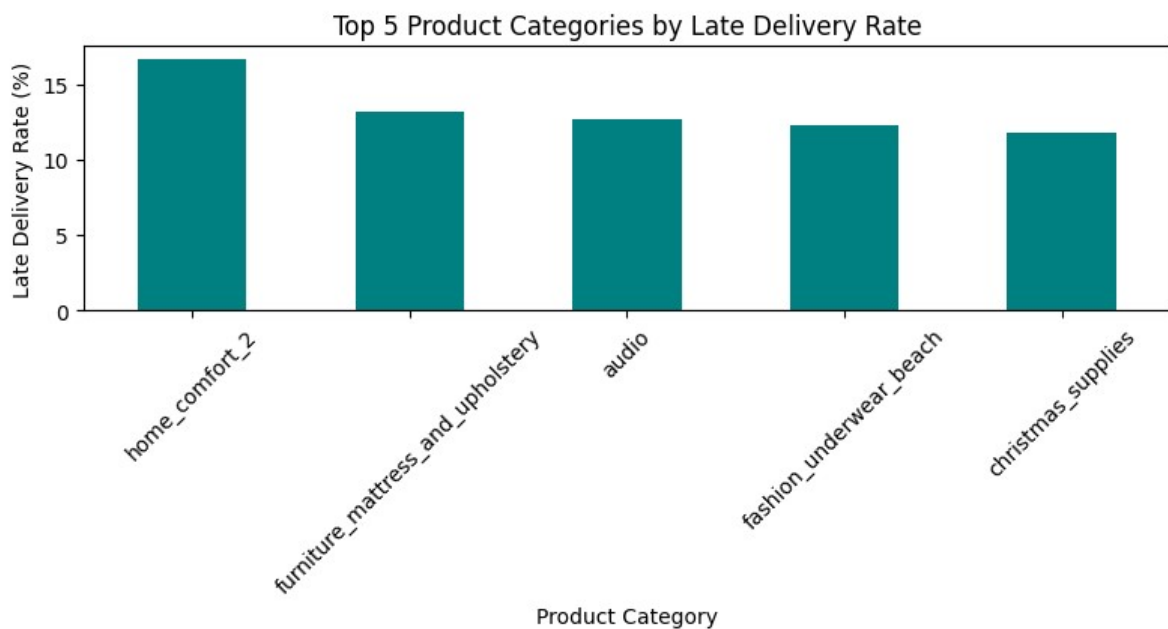
	Late_Flag
product_category_name_english	
home_comfort_2	16.666667
furniture_mattress_and_upholstery	13.157895
audio	12.637363
fashion_underwear_beach	12.213740
christmas_supplies	11.764706
books_technical	10.861423
home_comfort	10.138249
construction_tools_lights	9.868421

electronics	9.613300
--------------------	----------

food	9.607843
-------------	----------

dtype: float64

```
# Visualizing the Top 5
top5 = category_late_rate.head(5)
plt.figure(figsize=(8,5))
top5.plot(kind= 'bar',
          color='teal')
plt. title('Top 5 Product Categories by Late Delivery Rate')
plt.ylabel('Late Delivery Rate (%)')
plt.xlabel('Product Category')
plt.tight_layout()
plt.xticks(rotation= 45)
plt.show()
```



Candidate's Choice Challenge: Freight Cost vs Delivery Delay

This analysis examines whether higher shipping costs result in improved delivery performance.

```
## Bonus Analysis: Freight Cost vs Delivery Delay
master_df[['freight_value', 'Days_difference']].head()
```

	freight_value	Days_difference
0	8.72	7.0
1	22.76	5.0
2	19.22	17.0
3	27.20	12.0
4	8.72	9.0

```
master_df[['freight_value', 'Days_difference']].isnull().sum()
```

	0
freight_value	775
Days_difference	3229

```
dtype: int64
```

```
# Creating a dataset that contains only the columns needed
# for the bonus analysis
```

```
bonus_df = master_df[['freight_value', 'Days_difference']].dropna()
```

```
bonus_df.shape
```

```
(110196, 2)
```

```
# Visualizing the relationship between freight cost
# and delivery delay
```

```
plt.figure(figsize=(6,4))
```

```
plt.scatter(
    bonus_df['freight_value'],
    bonus_df['Days_difference'],
    alpha=0.3,
    color='teal'
)
```

```
plt.title('Freight Cost vs Delivery Delay')
plt.xlabel('Freight Cost')
```

```
plt.ylabel('Delivery Delay (Days)')

plt.tight_layout()
plt.show()
```



```
# Measuring the relationship between freight cost
# and delivery delay

correlation = bonus_df['freight_value'].corr(
    bonus_df['Days_difference']
)

print("Correlation:", correlation)
```

```
Correlation: 0.04024203581997367
```

```
# saving my final master_df
master_df.to_csv('final_logistics_dataset.csv', index=False)
```

```
from google.colab import files
files.download('final_logistics_dataset.csv')
```

```
# Final dataset shape
master_df.shape
```

(113425, 36)

```
# Final column names
master_df.columns
```

```
Index(['order_id', 'customer_id', 'order_status',
'order_purchase_timestamp',
      'order_approved_at', 'order_delivered_carrier_date',
      'order_delivered_customer_date', 'order_estimated_delivery_date',
      'customer_unique_id', 'customer_zip_code_prefix', 'customer_city',
      'customer_state', 'review_id', 'review_score',
'review_comment_title',
      'review_comment_message', 'review_creation_date',
      'review_answer_timestamp', 'Days_difference', 'Delivery_Status',
      'Late_Flag', 'order_item_id', 'product_id', 'seller_id',
      'shipping_limit_date', 'price', 'freight_value',
      'product_category_name', 'product_name_lenght',
      'product_description_lenght', 'product_photos_qty',
'product_weight_g',
      'product_length_cm', 'product_height_cm', 'product_width_cm',
      'product_category_name_english'],
      dtype='object')
```

```
# Missing values summary
master_df.isnull().sum().sort_values(ascending=False).head(10)
```

	0
review_comment_title	99920
review_comment_message	65497
Days_difference	3229
order_delivered_customer_date	3229
product_category_name_english	2402
product_photos_qty	2378
product_category_name	2378
product_name_lenght	2378
product_description_lenght	2378
order_delivered_carrier_date	1968

dtype: int64

Final Dataset Export

The final cleaned dataset was exported for dashboard development and further

The final cleaned dataset was exported for dashboard development and further business reporting.

Key Findings

1. Delivery delays vary significantly across states.
2. On-time deliveries receive substantially higher review scores.
3. Furniture-related products experience higher late-delivery rates.
4. Freight cost has almost no relationship with delivery delays ($r = 0.04$).
5. Logistics performance is a major driver of customer satisfaction.

Conclusion

The analysis demonstrates that delivery delays are a key contributor to negative customer reviews. Regional differences, product characteristics, and operational inefficiencies appear to influence delivery performance more strongly than shipping expenditure. Improving delivery reliability is likely to increase customer satisfaction and strengthen Veridi Logistics' overall service quality.